MIL-W-3970/5B
4 April 1977
SUPERSEDING
MIL-W-3970/5A
16 November 1964

MILITARY SPECIFICATION SHEET

WAVEGUIDE ASSEMBLIES, RIGID, 45-DEGREE E-PLANE BEND

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the waveguide assemblies described herein shall consist of this document and the latest issue of Specification MIL-W-3970.

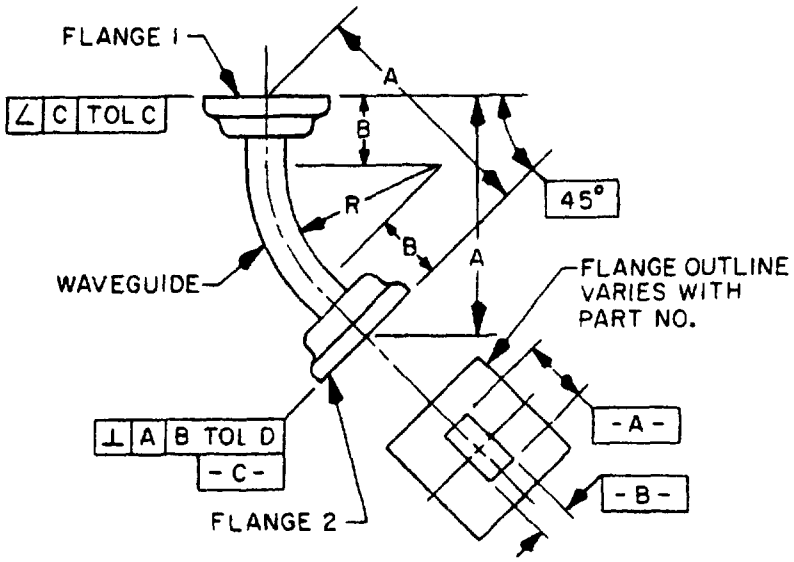


FIGURE 1. Assembly configuration.

FSC 5985

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:06Z
Check the source to verify that this is the current version before use.

TABLE I. Dash number, electrical and physical characteristics, and dimensions.

Dash No.	Waveguide M85/	Flange 1 M3922/	Flange 2 M3922/	Frequency range TE_{10} mode	Insertion loss max	VSWR max	Pressurization lb_f/in^2 gage	Dimensions 1/ 2/ 3/ A	B nom	C	D	R
				GHz	dB							
001	1-043	56-001	61-002	2.6 - 3.95	-	1.04	15	8.923 (222.64) ±.030 (.76)	1.811 (46.00)	.136 (3.45)	.012 (.030)	7.500 (190.50) ±.125 (3.18)
002	1-055	57-002	62-002	3.95 - 5.85	-	1.05	30	5.949 (151.10) ±.030 (.76)	1.207 (30.66)	.091 (2.31)	.008 (0.20)	5.000 (127.00) ±.125 (3.18)
003	1-067	55-001	60-001	5.85 - 8.20	-	1.05	30	4.196 (106.58) ±.025 (.64)	0.905 (22.99)	.067 (1.70)	.006 (0.15)	3.375 (85.72) ±.125 (3.18)
004	1-073	53-002	59-007	7.05 - 10	-	1.05	30	3.453 (87.71) ±.020 (.51)	0.754 (19.15)	.052 (1.32)	.005 (0.13)	2.751 (69.87) ±.063 (1.60)
005	1-079	53-001	59-006	8.2 - 12.4	-	1.04	30	3.081 (78.26) ±.020 (.51)	0.666 (16.92)	.046 (1.17)	.004 (0.10)	2.500 (63.50) ±.063 (1.60)
006	1-102	54-001	59-003	18 - 26.5	.11	1.05	30	1.628 (41.35) ±.015 (.38)	0.462 (11.73)	.025 (0.64)	.002 (0.05)	1.063 (27.00) ±.063 (1.60)
007	1-106	54-001	59-003	18 - 26.5	.11	1.05	30	1.628 (41.35) ±.015 (.38)	0.462 (11.73)	.025 (0.64)	.002 (0.05)	1.063 (27.00) ±.063 (1.60)
008	1-077	53-003	59-008	8.2 - 12.4	-	1.04	30	3.081 (78.26) ±.020 (.51)	0.666 (16.92)	.046 (1.17)	.004 (0.10)	2.500 (63.50) ±.063 (1.60)
009	1-071	53-004	59-009	7.05 - 10	-	1.05	30	3.453 (87.71) ±.020 (.51)	0.754 (19.15)	.052 (1.32)	.005 (0.13)	2.751 (69.87) ±.063 (1.60)
010	1-041	56-002	61-001	2.6 - 3.95	-	1.04	15	8.923 (226.64) ±.030 (.76)	1.811 (46.00)	.136 (3.45)	.012 (0.30)	7.500 (190.50) ±.125 (3.18)

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:06Z
Check the source to verify that this is the current version before use.

TABLE I. Dash number, electrical and physical characteristics, and dimensions - Continued.

Dash No.	Waveguide M85/	Flange 1 M3922/	Flange 2 M3922/	Frequency range TE_{10} mode	Insertion loss max	VSWR max	Pressurization lbf/in² gage	Dimensions 1/ 2/ 3/ A	B nom	C	D	R
				GHz	dB							
011	1-089	53-005	59-001	12.4 - 18	.07	1.05	30	2.339 (59.41) ±.015 (.38)	0.564 (14.33)	.036 (0.91)	.003 (0.08)	1.753 (44.53) ±.063 (1.60)
012	1-053	57-001	62-001	3.95 - 5.85	-	1.05	30	5.949 (151.10) ±.030 (.76)	1.207 (30.66)	.091 (2.31)	.008 (0.20)	5.000 (127.00) +.125 (3.18)
013	3-006	54-003	59-005	26.5 - 40	.15	1.08	30	1.191 (30.25) ±.010 (.25)	0.342 (8.69)	.019 (0.48)	.001 (0.03)	.763 (19.38) ±.063 (1.60)
014	1-065	55-002	60-002	5.85 - 8.20	-	1.05	30	4.196 (106.58) ±.025 (.64)	0.905 (22.99)	.067 (1.70)	.007 (0.18)	3.375 (85.72) ±.125 (3.18)
015	1-093	53-005	59-001	12.4 - 18	.07	1.05	30	2.339 (59.41) +.015 (.38)	0.564 (14.33)	.036 (0.91)	.003 (0.08)	1.753 (44.53) ±.063 (1.60)
016	1-103	54-002	59-004	18 - 26.5	.11	1.05	30	1.628 (41.35) ±.015 (.38)	0.462 (11.73)	.025 (6.35)	.002 (0.05)	1.063 (27.00) ±.063 (1.60)
017	3-006 3-008	54-003	54-003	26.5 - 40	.15	1.1	30	1.230±.015 (31.24)(.38)	.500 (12.70)	.012 (.30)	.001 (.03)	.500 (12.70)
018	3-006 3-008	67-005	67-005	26.5 - 40	.15	1.1	---	↓	↓	.014 (.36)	.001 (.03)	.500 (12.70)
019	3-010 3-012	67-006	67-006	33 - 50	.17	1.1	---			.014 (.36)	.001 (.03)	.500 (12.70)
020	3-014 3-016	67-007	67-007	40 - 60	.19	1.1	---			.014 (.36)	.001 (.03)	.500 (12.70)
021	3-017 3-019	67-008	67-008	50 - 75	.21	1.1	---			.012 (.30)	.001 (.03)	.500 (12.70)
022	3-020 3-022	67-009	67-009	60 - 90	.23	1.1	---			.012 (.30)	.001 (.03)	.500 (12.70)

1/ Dimensions are in inches.
2/ Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3/ Metric equivalents are in parentheses.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:06Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Dimensions and configuration: See figure 1 and table I.

Electrical characteristics: See table I.

Part number: M3970/5-(dash number from table I).

Cross reference: See table II.

TABLE II. Cross reference.

Part number M3970/5-	Type designation	Part number M3970/5-	Type designation
001	UG-728/U	009	UG-736/U
002	UG-729/U	010	UG-737/U
003	UG-730/U	011	UG-738/U
004	UG-731/U	012	UG-739/U
005	UG-732/U	013	UG-740/U
006	UG-733/U	014	UG-741/U
007	UG-734/U	015	UG-742/U
008	UG-735/U	016	UG-743/U

NOTE: Assemblies covered by this specification are substitutable for the following commercial types:

Part number M3970/5-	Manufacturer's code	Commercial type
017	73293	44321H-451
018	73293	44322H-451
019	73293	44323H-451
020	73293	44324H-451
021	73293	44325H-451
022	73293	44326H-451

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MI,
Navy - EC, OS
Air Force - 11, 17, 99
DSA - ES

User activities:
Army - MU
Navy - MC, CG, AS
Air Force - 14, 19

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5985-0835-2)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:06Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No 22-R255

INSTRUCTIONS The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government Fold on lines on reverse side, staple in corner, and send to preparing activity Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements Attach any pertinent data which may be of use in improving this document If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:06Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D.C. 20360

POSTAGE AND FEES PAID
NAVY DEPARTMENT
DOD 316

OFFICAL BUSINESS

COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:06Z
Check the source to verify that this is the current version before use.